

Xiling Li

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Personal Website: <https://xilinggrantli.github.io> Google Scholar DBLP

RESEARCH INTERESTS	My research focuses on secure and private data management and its applications including verifiable query evaluation, oblivious outsourced querying and privacy-preserving machine learning, and works extensively with secure multiparty computation, zero-knowledge proofs and differential privacy.	
EDUCATION	Ph.D. Computer Science , Northwestern University • Advisor: Dr. Jennie Rogers	Jun 2021 - Present
	M.S. Computer Science , University of Washington • Advisor: Dr. Martine De Cock • Thesis: <i>Privacy-Preserving Filter-based Feature Selection with Secure Multiparty Computation</i>	Dec 2020
	B.S. Computer Science , University of California, San Diego	Dec 2016
RESEARCH EXPERIENCE	Research Assistant , Northwestern University • ZKSQL (VLDB 2023) - 40+ Citations: Proposed the first work [2] on verifiable and efficient query evaluation with zero knowledge proofs for ad-hoc SQL queries in an operator-at-a-time fashion. • RESCU-SQL (VLDB 2023 demo): Proposed the first pragmatic OLAP system [1] with all-but-one malicious security for ad-hoc SQL queries.	Jun 2021 - Present
	Research Intern , Dolby @ATG Experience Delivery • Prototyped scalable multimodal lakehouse and upstream data pipeline for heterogeneous AI workloads.	Sep 2025 - Dec 2025
	Research Assistant , University of Washington @PPML Group • ICML 2021 - 70+ Citations: Proposed the first general secure multiparty computation protocol [3] for FFS with malicious security and honest majority, and instantiated feature score protocol with MS-GINI. • UbiTtention 2020 Workshop (UbiComp-ISWC 2020): Proposed Mean-Split Gini Impurity algorithm (MS-GINI) [4] for Filter-based Feature Selection (FFS).	Sep 2019 - May 2021
TEACHING EXPERIENCE	Guest Lecturer • <i>Database Architecture and Query Evaluation</i> , COMP_SCI 339, Northwestern University Fall 2023 • <i>Relational Algebra</i> , COMP_SCI 339, Northwestern University Spring 2024	
	Teaching Assistant • <i>COMP_SCI 339: Intro to Database Systems</i> , Northwestern University	Spring 2023-2025
INDUSTRIAL EXPERIENCE	Data Scientist , IBM @Watson IoT • Built a knowledge graph with Neo4J for a reasoning module in a disaster prevention platform. • Integrated an image pre-processing module with OpenCV into a vision system for manufacturing. • Developed a millisecond-level face recognition module using FaceNet for warehouse management.	Jan 2018 - Aug 2019
	Android Developer , Shenzhen Das Intellitech Co., Ltd @R&D Department • Developed an Android module for door access control by voice within an intelligent building system.	Jul 2017 - Dec 2017
SERVICES	Reviewer: ICML 21-25; NeurIPS 21-25; ICLR 22-26; AISTATS 25-26; AAAI 26; Asiacrypt 25	
TECHNICAL SKILLS	C++, Python, EMP-toolkit , PyTorch, MP-SPDZ , AWS EC2, Ubuntu, Docker	
OPEN SOURCE ARTIFACTS	Xiling Li , Chenkai Weng, Yongxin Xu, Xiao Wang, Jennie Rogers. <i>ZKSQL: Verifiable and Efficient Query Evaluation with Zero-Knowledge Proofs</i> . https://github.com/vaultdb/zksql , Feb 2023.	

INVITED
TALKS

Efficient Oblivious Database Joins, *Northwestern Database Reading Group*, May 2025.
ZKSQL, *Northwestern Database Reading Group*, Jan 2025.
Privacy + Machine Learning, *Northwestern AI Journal Club*, Nov 2021.

SELECTED
PUBLICATIONS

- [1] **Xiling Li***, Gefei Tan*, Xiao Wang, Jennie Rogers, Soamar Homs. *RESCU-SQL: Oblivious Querying for the Zero Trust Cloud*. In Proceedings of the VLDB Endowment (PVLDB), Volume 16, No. 12, 4086-4089, 2023. DOI: [https://doi.org/ 10.14778/3611540.3611627](https://doi.org/10.14778/3611540.3611627).
- [2] **Xiling Li**, Chenkai Weng, Yongxin Xu, Xiao Wang, Jennie Rogers. *ZKSQL: Verifiable and Efficient Query Evaluation with Zero-Knowledge Proofs*. In Proceedings of the VLDB Endowment (PVLDB), Volume 16, No. 8, 1804-1816, 2023. DOI: <https://doi.org/10.14778/3594512.3594513>.
- [3] **Xiling Li**, Rafael Dowsley, Martine De Cock. *Privacy-Preserving Feature Selection with Secure Multiparty Computation*, In Proceedings of the 38th International Conference on Machine Learning, PMLR 139:6326-6336, 2021.
- [4] **Xiling Li**, Martine De Cock. *Cognitive load detection from wrist-band sensors*. In Adjunct Proceedings of the 2020 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2020 ACM International Symposium on Wearable Computers (UbiComp-ISWC '20). ACM, New York, NY, USA, 456–461. DOI: <https://doi.org/10.1145/3410530.3414428>.